Weight-Sharing Order Changes Linear Expressivity

Separation and Commutator Bounds



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CENTRAL IDEA Same parameter count. Same reuse frequency. Different representable maps. Commutators quantify what happens when fixed weights are reordered.

01 Motivation & research question

Does layer order matter when two networks reuse exactly the same number of weight matrices?

ALTERNATING

Execution order → **A** → **B** → **A** → **B**

$$T_{ABAB} = BABA = (BA)^2$$

BLOCK

Execution order → **A** → **A** → **B** → **B**

$$T_{AABB} = B^2A^2$$

2 real matrices **4 layers** **Reuse: (2, 2)**

Both schedules have $2n^2$ trainable parameters.

Answer: yes. For every $n \geq 2$, block sharing can represent real linear maps that alternating sharing cannot.

02 Exact expressivity separation

For real n × n matrices, n ≥ 2:

$$\{(BA)^2\} \subsetneq \{B^2A^2\}$$

A concrete witness in two dimensions

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \mathrm{diag}(1, 2)$$

$$B^2A^2 = D = \mathrm{diag}(-1, -4)$$

1. Every alternating map is a real matrix square. Every such square is also attainable by block sharing (take A = I).
2. If $C^2 = D$, then C commutes with D. Thus C preserves the one-dimensional -1 eigenspace.
3. On that eigenspace, a real scalar would have to square to -1. This is impossible: D has no real square root.

Extend A and B with identity blocks for n > 2.
The separation holds even with invertible weights.

03 Fixed weights: sensitivity to order

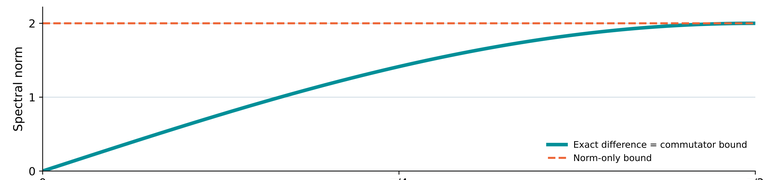
An adjacent swap contributes a commutator, scaled by the factors surrounding it.

$$\Delta T = P[W_j, W_i]Q, \qquad [A, B] = AB - BA$$

$$\|T_\sigma - T_\tau\| \leq \min_\gamma \sum_{r \in \gamma} \|P_r\| \, \|[W_{j_r}, W_{i_r}]\| \, \|Q_r\|$$
$$\leq M^{L-2} C_W(\sigma, \tau)$$

M = max weight norm; C_W is the minimum sum of commutator norms along adjacent-swap paths. All norms are spectral.

An exactly attainable regime



Analytic example: A = diag(1, −1), B = R(θ), identity exterior factors. Both commutator bounds equal $2|\sin\theta|$; the norm-only bound is 2.

04 Full-data evaluation

MNIST + Fashion-MNIST: 60,000 train / 10,000 test each.
Training-only PCA: n = 16, 32 · 4 schedules · 10 seeds.
160 fits; AABB / ABAB use matched initial weights.

Test accuracy (%) · mean ± sample SD, ten seeds

Dataset	n	AABB	ABAB
MNIST	16	77.550 ± 0.000	77.556 ± 0.008
MNIST	32	83.600 ± 0.000	83.560 ± 0.000
Fashion-MNIST	16	71.520 ± 0.000	72.008 ± 0.033
Fashion-MNIST	32	75.960 ± 0.000	76.034 ± 0.011

80 / 160 fits reached the 10,000-iteration cap.
All AAAA and ABAB fits hit the cap; all AABB and ABCD fits satisfied stopping criteria. These are finite-budget outcomes.

Bias-free linear maps · L-BFGS-B · MSE with $10^{-4}\|T\|_F^2$ penalty.
The results do not establish a classification advantage.

05 How useful are the bounds?

80 learned AABB / ABAB models × 15 pairs of reorderings
= **1,200 dependent comparisons**.

77.5% — **Uniform bound**
930 / 1,200 comparisons
Improves the norm-only baseline $2M^4$

7.7% — **Transported bound**
92 / 1,200 comparisons
Improves the tighter baseline $\|T\sigma\| + \|T\tau\|$

Different baselines; not a head-to-head win rate.
These dependent comparisons are descriptive, not independent trials.

06 Interpretation, scope & next steps

REPRESENTABILITY
A strict result for real, deep linear networks and full linear maps. Equal parameter counts need not imply equal expressivity.

GEOMETRY & PERFORMANCE
Different maps need not change output Gram geometry. With orthogonal factors, $X^TT^TTX = X^TX$ for every ordering.

OPTIMIZATION & SCOPE
Optimization failure is not proof of nonrepresentability. Classification gains and results for nonlinear LLMs require separate evidence.

Report these three questions separately.

The takeaway

Sharing order is an architectural choice.
It can change the set of representable linear maps. Commutators certify sensitivity to reordering fixed weights. Neither result alone establishes better predictive performance.

Selected references

1. Takase & Kiyono. Lessons on parameter sharing across layers in transformers. SustaiNLP, 2023.
2. Gallier. Logarithms and square roots of real matrices. arXiv:0805.0245.
3. Childs et al. Theory of Trotter error with commutator scaling. Physical Review X 11:011020, 2021.